

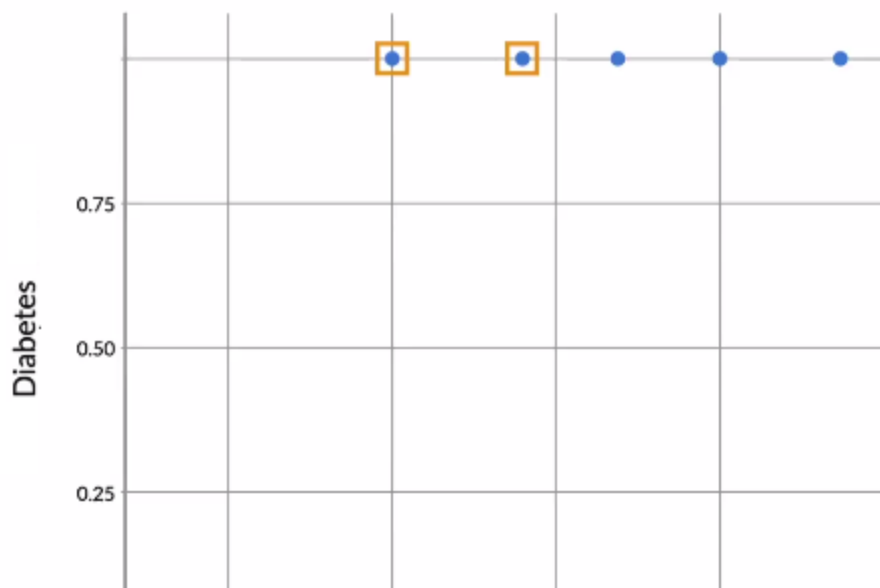
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Summary

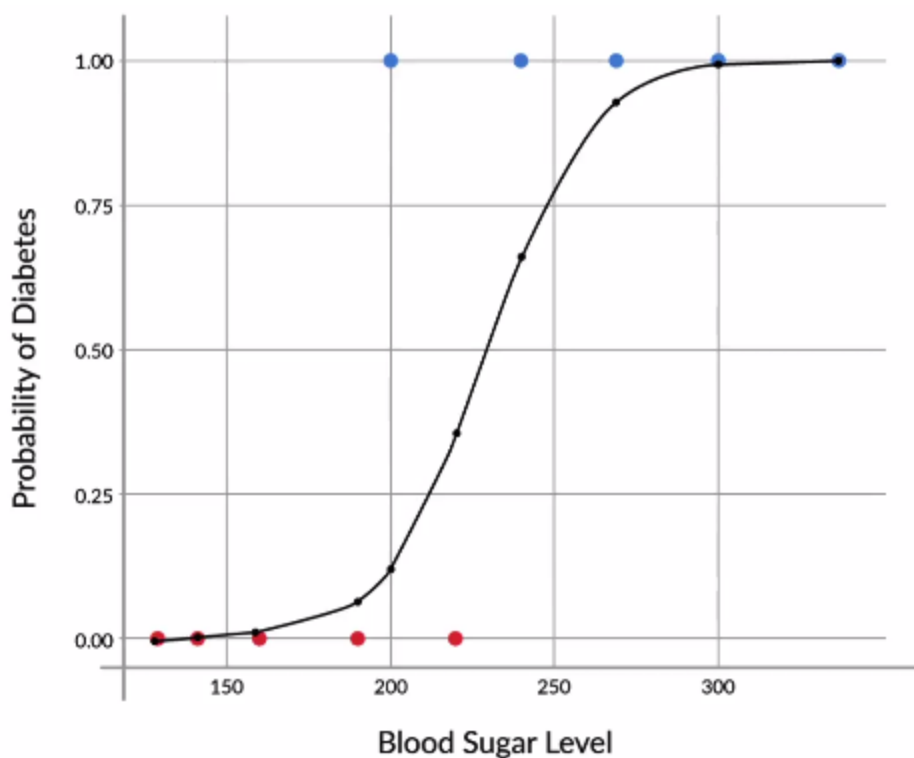
You first learnt what a **binary classification** is. Basically, it is a classification problem in which the target variable has only 2 possible values.

You then went through the **diabetes example** in detail, wherein you tried to predict whether a person has diabetes or not based on that person's blood sugar level.

You saw why a **simple boundary decision approach** does not work very well for this example. It would be too risky to decide the class blatantly on the basis of the cutoff because, especially in the middle, the patients could belong to any class – diabetic or non-diabetic.



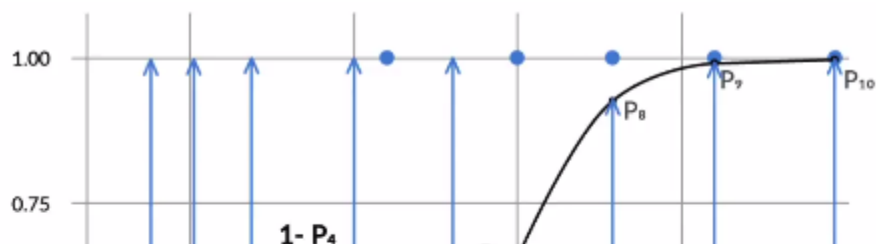
Hence, you learnt that it is better to talk in terms of **probability**. One such curve which can model the probability of diabetes very well is the **sigmoid curve**.



Its equation is given by the following expression:

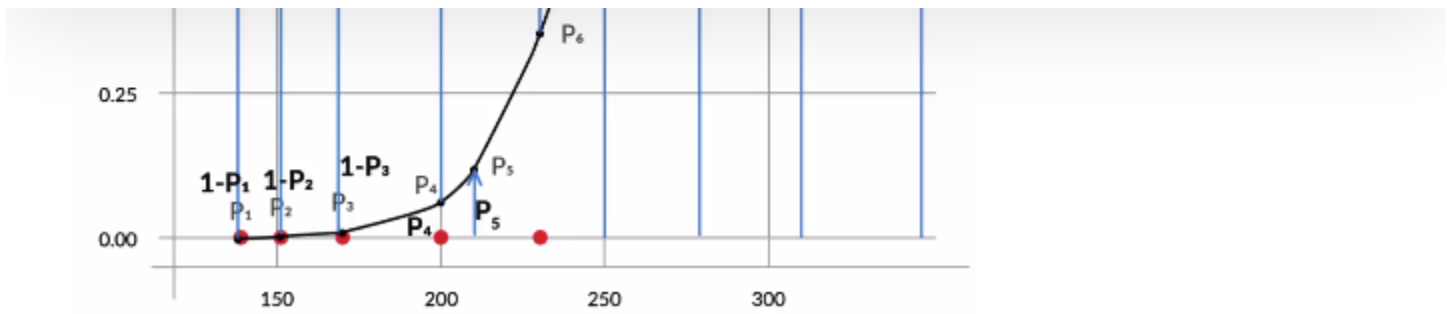
$$P(\text{Diabetes}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Then, you learnt that in order to find the **best-fit sigmoid curve**, you need to vary β_0 and β_1 until you get the combination of beta values that maximises the **likelihood**. For the diabetes example, the likelihood is given by the expression:



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$$Likelihood = 1 - P_1 1 - P_2 1 - P_3 1 - P_4 P_5 1 - P_6 P_7 P_8 P_9 P_{10}$$

It is the product of:

$$[(1-P_i)(1-P_i) \text{ ----- for all non-diabetics -----}] * [(P_i)(P_i) \text{ ----- for all diabetics -----}]$$

This process, where you vary the betas until you find the best fit curve for the probability of diabetes, is called **logistic regression**.

After this, you saw a simpler way of interpreting the equation for logistic regression. You saw that the following linearised equation is much easier to interpret:

$$\ln \frac{p}{1-p} = \beta_0 + \beta_1 x$$

The left-hand side of this equation is what is called **log odds**. Basically, the **odds** of having diabetes ($P/(1-P)$), indicate how much likelier a person is to have diabetes than to not have it. For example, a person for whom the odds of having diabetes are equal to 3, is 3 times more likely to have diabetes than to not have it. In other words, $P(\text{Diabetes}) = 3 * P(\text{No diabetes})$.

You also saw how odds vary with variation in x . Basically, with every **linear increase** in x , the increase in odds is **multiplicative**. For example, in the diabetes

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